Junyi Chu

Postdoctoral Scholar

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Appointments

2024-present *Postdoctoral Scholar*, Psychology, Stanford University
2023-2024 *Postdoctoral Scholar*, Psychology, Harvard University

Education

2017-2023 PhD in Cognitive Science, Massachusetts Institute of Technology (Cambridge, MA)
2011-2015 B.S. in Child Development & Cognitive Studies, Vanderbilt University (Nashville, TN)

Honors & Awards

2025 Postdoctoral Fellow, Stanford Institute for Human-Centered Artificial Intelligence (HAI)
2024 Disciplinary Diversity & Integration Award for symposia, Cognitive Science Society
2022 Fellow, Diverse Intelligences Summer Institute
2022 Diversity, Equity, Inclusion, and Justice Impact Award, MIT Department of Brain and Cognitive Sciences
2019 Angus MacDonald Award for Excellence in Undergraduate Teaching, MIT
2018 Henry E. Singleton (1940) Fellowship, MIT
2017 Presidential Graduate Fellowship, MIT (top 15% of class)
2015 Best Undergraduate Thesis in Cognitive Studies, Vanderbilt University
2014 Littlejohn Summer Research Fellowship, Vanderbilt University
2011-2015 Dean's Achievement Award, Peabody College at Vanderbilt University (full tuition scholarship)

Research Grants

2025-2026 Learning through Creation with Generative AI Seed Grant
  Source: Stanford Accelerator for Learning and Stanford Institute for Human-Centered Artificial Intelligence (HAI)
  Title: Enhancing math learning and engagement through game creation
  Role: co-PI, w/ Judith Fan, Hyo Gweon, Nick Haber, & Hari Subramonyam
  Amount: $25,000
2025-2026 Joyful Learning Seed Grant
  Source: Stanford Accelerator for Learning
  Title: Joyful Self-Surprise as an Engine for Learning in Early Childhood
  Role: Co-I, w/ Adani Abutto, Judith Fan, & Hyo Gweon
  Amount: $25,000

2025-2026 Rand Innovation Fund
Source: Harvard University Department of Psychology
Title: Creative Goal Generation and Evaluation in Humans and Machines
Role: co-PI, w/ Tomer Ullman & Jennifer Hu
Amount: $6,000

2025-2026 Ditmars Innovation Fund
Source: Harvard University Department of Psychology
Title: Play, a computational perspective
Role: co-I, w/ Tomer Ullman
Amount: $30,000

Journal articles

2025 **Chu, J.**, Rule, J., Goddu, M., Pinter, V., Reagan, E.R., Bonawitz, E., Gopnik, A., & Ullman, T.D. (2025) Children selectively manipulate task difficulty when "playing for fun" vs. "trying to win", *Developmental Psychology*

2024 **Chu, J.**, Tenenbaum, J.B., & Schulz, L.E. (2024) In praise of folly: Flexible goals and human cognition. *Trends in Cognitive Sciences*

2023 **Chu, J.** & Schulz, L.E. (2023) Not playing by the rules: Exploratory play, rational action, and efficient search. *Open Mind* 7: 294–317.

2023 Erel, Y., Adams Shannon, K., **Chu, J.**, Scott, K., Kline Struhl, M., Cao, P., Tan, X., Hart, P., Raz, G., Piccolo, S., Mei, C., Potter, C., Jaffe-Dax, S., Lew-Williams, C., Tenenbaum, J., Fairchild, K., Bermano, A.., Liu, S. (2023). iCatcher+: Robust and automated annotation of infant gaze from videos collected in laboratory, field, and online studies. *Advances in Methods and Practices in Psychological Science*

2021 **Chu, J.** & Schulz, L.E. (2021) Children selectively endorse speculative conjectures. *Child Development.*

2020 **Chu, J.** & Schulz, L.E. (2020) Play, Curiosity, and Cognition. *Annual Review of Developmental Psychology*, 2

2020 **Chu, J.**, Cheung, P., Schneider, R., Sullivan, J. & Barner, D. (2020) Counting to infinity: Does learning the syntax of the count list predict knowledge that numbers are infinite? *Cognitive Science*, 44:e12875.

2019 Wagner, K., **Chu, J.**, & Barner, D. (2019) Do children's number words begin noisy? *Developmental Science, 22*(1):e12752.

2017 Barner, D., Athanasopoulou, A., **Chu, J.**, Lewis, M., Marchand, E., Schneider, R., & Frank, M. (2017). A one-year classroom-randomized trial of mental abacus instruction for first- and second- grade students. *Journal of Numerical Cognition, 3*(3)

2017 Scott, K.M., **Chu, J.**, and Schulz, L.E. (2017). Lookit (Part 2): Assessing the viability of online developmental research: Results from three case studies. *Open Mind, 1*(1), 15-29.

2017 **Chu, J.**, Rittle-Johnson, B and Fyfe, E.R. (2017). Diagrams benefit symbolic problem solving. *British Journal of Educational Psychology, 87*, 273-287.

Manuscripts under review, in revision, or in preparation

under review Rule, J., **Chu, J.**, Goddu, M., Pinter, V., Reagan, E.R., Bonawitz, E., Gopnik, A., & Ullman, T.D. *Children's play differs from both exploring and exploiting.*

in prep **Chu, J.** & Schulz, L.E. *Because I want to: Valuing goals for their own sake.*

in prep **Chu, J.**, Cheyette, S., Digges-Galligan, S., Tenenbaum, J.B., & Schulz, L.E. *Curious, creative, and complex: Measuring goal generation in play.*

Peer-reviewed Conference Proceedings

[+] mentees for whom I served as primary or co-mentor.

2025 **Chu, J.**, Zheng, K.[+] , *&* Fan, J.E. (2025). What makes people think a puzzle is fun to solve? *Proceedings of the 47th Annual Conference of the Cognitive Science Society.*

2025 **Chu, J.**, O'Keeffe, M.[+], Liu, S., Bonawitz, E., *&* Ullman, T.D. (2025). Stumped! Learning to think outside the box in 3–7-year-olds. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.*

2025 Todd, G.[+], **Chu, J.**, Davidson, G., *&* Xu, W. (2025). Novel goal creation and evaluation in open-ended games. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.*

2025 Davidson, G., Todd, G., Colas, C., **Chu, J.**, Togelius, J., Tenenbaum, J.B., Gureckis, T.M., *&* Lake, B. (2025). Goal inference using reward-producing programs in a novel physics environment. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.*

2025 Collins, K.*, Todd, G.*, Zhang, C.E., Weller, A., Togelius, J., **Chu, J.**, Wong, L., Griffiths, T., *&* Tenenbaum, J.B. (2025). Generation and evaluation in the human invention process through the lens of game design. *Proceedings of the 47th Annual Conference of the Cognitive Science Society.*

2024 **Chu, J.**, Hu, J., *&* Ullman, T.D. (2024). The task task: Creative problem generation in humans and language models. *Proceedings of the 46th Annual Conference of the Cognitive Science Society.*

2022 **Chu, J.** *&* Schulz, L.E. (2022). Because I want to: Valuing goals for their own sake. *Proceedings of the 44th Annual Conference of the Cognitive Science Society.*

2022 Coates, N., Siegel, M., **Chu, J.**, Kline, M., Tenenbaum, J.B., *&* Schulz, L.E. (2022). Preschoolers' sensitivity to abstract correlations in the properties of sets and functions. *Proceedings of the 44th Annual Conference of the Cognitive Science Society.*

2021 Diggs-Galligan, S.[+], **Chu, J.**, Tenenbaum, J.B., *&* Schulz, L.E. (2021). Explore, exploit, create: Inventing goals in play. *Proceedings of the 43rd Annual Conference of the Cognitive Science Society.*

2020 **Chu, J.** *&* Schulz, L.E. (2020). Exploratory play, rational action, and efficient search. *Proceedings of the 42nd Annual Conference of the Cognitive Science Society.*

2019 **Chu, J.**, Gauthier, J., Levy, R., Tenenbaum, J.B., *&* Schulz, L.E. (2019). Query-guided visual search. *Proceedings of the 41st Annual Conference of the Cognitive Science Society.*

2018 **Chu, J.** *&* Schulz, L.E. (2018). Cognitive pragmatism: Children flexibly endorse facts and conjectures. *Proceedings of the 40th Annual Conference of the Cognitive Science Society.*

2016 **Chu, J.**, Wagner, K., *&* Barner, D. (2016). Children learn non-exact number word meanings first. *Proceedings of the 38th Annual Conference of the Cognitive Science Society.*

2015 **Chu, J.**, Fyfe, E.R., *&* Rittle-Johnson, B. (2015). Diagrams benefit symbolic problem-solving. *Proceedings of the 37th Annual Meeting of the Cognitive Science Society.*

Talks and Presentations

Invited Talks & Seminars

2027 Title TBD
Institute for Advanced Study in Toulouse & Toulouse School of Economics

2026 The Generative Mind: Cognitive Foundations of Play and Discovery
Department of Psychology, Williams College

2026 Playful problem solving
Legos & Learn, Stanford School of Medicine, Office of Community Engagement in Anesthesiology

2026 Designing for play
Cooperative Systems & Intelligence Lab, National University of Singapore

2026 The Generative Mind: Cognitive Foundations of Play and Discovery
Department of Psychology, Emory University

2025 Playful problem solving
ARC Prize Foundation: Measuring Intelligence with Interactive Benchmarks, MIT

2025 Playful problem selection
CogSci Workshop on Meta-reasoning: deciding which game to play, which problem to solve, and when to quit.

2024 The puzzle and promise of play
Developmental Psychology Brown Bag, Brown University

2023 Goals, Play, and Cognitive Pragmatism: A study of flexible human minds
Developmental Psychology Brown Bag, Stanford University

2023 In praise of folly: goals, play, and problem-making
Intrinsically Motivated Open-ended Learning (IMOL) Conference, Sorbonne University

2023 In praise of folly: Flexible goals and intrinsic motivation
Affective Brain Lab, MIT

2022 Goals, play, and cognitive pragmatism
Computation, Cognition, and Development Lab, Harvard University

2022 Goals, play, and cognitive pragmatism
Shenhav Lab, Brown University

2022 Goals, play, and cognitive pragmatism
Cognitive Tools Lab, UC San Diego

2022 Goals, play, and cognitive pragmatism
Cognition and Learning Center, Rutgers University

2022 Goals, play, and cognitive pragmatism
Early Learning & Cognition Lab, UC San Diego

2021 Goals, Play, and Cognitive Pragmatism
Bonawitz Computational Cognitive Development Lab, Harvard University

2020 Play, Curiosity, and Cognition: How our goals shape learning and thinking
Social Learning Lab, Stanford University

2019 Beyond accuracy and efficiency: How goals shape learning and exploration
Emerging Scholars of Psychological Science, Princeton University

Refereed Conference Abstract Presentations

2026 Klotz, M., **Chu, J**., Liquin, E. (2026, April). *Step-by-step or all-at-once: Preference for insight problem solving across goals and age groups.* Poster presented at the biennial meeting of the Cognitive Development Society, Montréal, Canada.

2026 Liu, Y., Crighton, F. E., **Chu, J**., Liquin, E., Bonawitz, E., *&* Mills, C. M. (2026). *Learning who helps about what: Variability in children's information-seeking* Poster presented at the Origins of the Social Mind (OSM) Preconference, Society for Personality and Social Psychology (SPSP), Chicago, IL

2025 Liu, S., **Chu, J**., Bonawitz, E., *&* Ullman, T.D. (2025, July). *Stumped! Learning to think outside the box in 3–7-year-olds.* Poster presented at Society for Research in Child Development (SRCD) Biennial Meeting, Minneapolis, MN.

2024 Wong, M., **Chu, J**., Ullman, T., *&* Bonawitz, E. (2024, March). *Objects to think by: What predicts children's pretend play preferences?* Poster presented at the biennial meeting of the Cognitive Development Society, Pasadena, CA.

2024 Mittal, A., **Chu, J**., *&* Schulz, L.E. (2024, March). *Because I want to: Valuing goals for their own sake.* Poster presented at the biennial meeting of the Cognitive Development Society, Pasadena, CA.

2024 Wang-Zhao, J., **Chu, J**., Bonawitz, E., *&* Ullman, T.D. (2024, March). *Sensible nonsense: Children's explanations of physical violations vary by age.* Poster presented at the biennial meeting of the Cognitive Development Society, Pasadena, CA.

2023 **Chu., J.**, Cheyette, S., Tenenbaum, J.B., *&* Schulz, L.E. (2023, May) *Curious, creative, and complex: goal generation in play.* Poster presented at the Curiosity, Creativity, and Complexity Conference, New York, NY.

2022 Riskin, S., **Chu, J.**, *&* Schulz, L.E. (2022, July) *Do preschoolers engage in rational reconsideration?.* Poster presented at the biennial meeting of the Cognitive Development Society, Madison, WI.

2021 Lapidow, E., **Chu, J.**, *&* Walker, C.M. (2021, July) *Knowing the shape of the solution: Causal structure constrains evaluation of possible causes..* Poster presented at the 43rd Annual Meeting of the Cognitive Science Society (virtual).

2019 Brooke-Wilson, T., Rosenfeld, J.S., Hofer ,M., **Chu, J.**, *&* Tenenbaum, J. (2019, July) *Simplicity and probability in human judgment.* Poster presented at the 41st Annual Meeting of the Cognitive Science Society, Montreal, Canada.

2019 **Chu, J.**, Gauthier, J., Levy, R., Tenenbaum, J.B., *&* Schulz, L.E. (2019, July) *Query-guided visual search.* Poster presented at the 41st Annual Meeting of the Cognitive Science Society, Montreal, Canada.

2014 **Chu, J.**, Hall, E., Loehr, A., *&* Rittle-Johnson, B. (2014, November) *Promoting mathematical problem solving and explanation via homework.* Poster presented at the Tennessee Psychological Association Annual Convention, Nashville, TN.

Teaching

Guest Lectures

2026 PSYCH 230: Computational Models of the Creative Process (Stanford)
2026 PHIL 14N: Philosophy of Games (Stanford)
2024 EDU H128: Curiosity in Learning and Development (Harvard Graduate School of Education)
2018 9.85: Infant and Early Childhood Cognition (MIT)

Graduate Teaching Assistant, *MIT*

2019 9.46 Neuroscience of Morality
2018 9.85 Infant and Early Childhood Cognition

Undergraduate Teaching Assistant, *Vanderbilt University*

2015 PSY1500, *Introduction to Cognitive Development*

Other Teaching

2018-2020 AP Physics *&* SAT Mathematics. MIT Academic Teaching Initiative
Instructor and Co-Director. Provided free weekly lectures, problem sets, and mock exams for 200+ students

Mentoring Activities

Stanford, Department of Psychology

Master's Kalu Obasi (2026–present; Master's thesis in Symbolic Systems); Connie Chang (2026–present; visiting from Harvard Graduate School of Design)
Undergrad. Imogen Lee (2025–present), James Zhang (2025–present), Carey Chang (2025–2026)

Harvard, Department of Psychology / Graduate School of Education

Master's — Yujung Janie Ro (2024–2025), Jiayi Song (2024), Silvia Liu (2023–2025; *now PhD student at* UW Madison), Wenxiu Wang (2023)

Undergrad. — Misha O'Keefe (2024–2025; *now PhD student at* NYU), Julio Caggiano (2024–2025), Kacper Malinowski (2024), Miranda Zhang (2024), Sanghee Song (2023–2024; *now PhD student at* Harvard), Verity Pinter (2023–2024), Eleanor Gao (2023–2024), Juliana Goldsby (2023–2024)

MIT, Department of Brain and Cognitive Sciences

Undergrad. RAs (n=32) — [*]Successfully applied for non-lab research funding or fellowships. [†]Student from the MIT Summer Research Program, an internship for non-MIT undergraduates from groups underrepresented in STEM.

Alex Taylor (2023–2024), Yuka Machino[*] (2023; *now PhD student at* Stanford), Melanie Albanese (2023), Ana de la Vega (2023), Felicia Du (2023), Michael Sheehan (2023), Olivia Joseph (2023), Nunu Lakew (2023), Fedaa Alsoufi (2023), Katherine Zeng (2023), Bianca Santi[*] (2022–2023; *now PhD student at* Princeton), Kameron Garland (2022–2023), Lauren Keller (2022), Beyza Ciftci (2022; *now PhD student at* CEU), Grace Zhang (2022), Faith Choe (2022), Hyunjin Lee (2022), Ashley Lederman[*] (2022), Christopher Montejo (2021–2022; FIU), Sofia Riskin[*] (2021–2023; Smith College), Sophia Diggs-Galligan[*] (2020–2022), Kailande Cassamajor[†] (2020), Naomi Kirimi[*] (2019–2020), Jinger Chong[*] (2019), Heidi Li[*] (2019), Grace Cowles (2019), Lucy Fu (2019), Gabriel Kane (2019), Alison Plump (2019; Dartmouth), Rucha Kelkar[*] (2018–2019), Jaemarie Solyst[†] (2018; *now PhD student at* CMU), Cindy Zhou (2018; Wellesley)

High School Interns (n=14) — *Established a partnership with Somerville High School & Black Girls Code to provide summer internships for high school students from groups underrepresented in STEM. Mentored students through the full research lifecycle (video annotation, stimuli design, participant recruitment, and data processing); facilitated weekly journal clubs and professional development seminars; supervised original research proposals culminating in a capstone poster session.*

Asmita Mittal, Andre Weiss, Salina Musyaju, Crystal Liu, Kevin Wen, Zoe Price, Liora Jones, Katherine Johnson, Leensyn Rivera, Lillian Switkes, Monica Correia, Claire Ma, Cynthia Lei, Elisa Dimagiba

Other Mentoring

2025 — Mentor, Stanford Science Small Groups
2024 — Mentor, Stanford Paths to PhD
2024-present — Mentor, Cognitive Science Society Mentorship Program
2023-present — Mentor, Cognitive Development Society Mentorship Program
2023 — Mentor, Harvard PPREP (Prospective PhD & RA Event in Psychology)
2019-2022 — Mentor and Organizer, MIT Brain & Cognitive Sciences Application Assistance Program

Professional Service

Service to the University and Broader Community

2023-2025 — ManyBabies Teaching, Training, and Open Science committee
2022-2023 — Lookit Researcher Working and Development Group
2022 — MIT Brain & Cognitive Sciences Institute Visiting Committee
2021-2022 — MIT Brain & Cognitive Sciences Graduate Affairs Committee
2020-2021 — Somerville High School Beyond the Classroom Learning Committee

Service to the Field

Workshops & Symposia Organized

2025 **Chu, J.**, Verma, A., Davidson, G., Fraser, R., & Fan, J. (2025, July). *Minds in the Making: Cognitive Science and Design Thinking.* Workshop at the 47th annual meeting of the Cognitive Science Society. minds-making.github.io

2024 Colas, C., Ady, N., **Chu, J.**, Sancaktar, C., Teodorescu, L., Davidson, G., & Molinaro, G. (2024, December). *Intrinsically Motivated Open-ended Learning.* Workshop at NeurIPS. imol-workshop.github.io

2024 Colas, C., Molinaro, G., Hawkins, R., & Chu, J. (2024, July). *What Should I Do Now? Goal-Centric Outlooks on Learning, Exploration, and Communication.* Symposium at the 46th Annual Meeting of the Cognitive Science Society.

2023 Colas, C., Teodorescu, L., Ady, N., Sancaktar, C., & Chu, J. (2023, December). *Intrinsically Motivated Open-ended Learning.* Workshop at NeurIPS, New Orleans, LA. imol-workshop.github.io

2021 **Chu, J.** & Schulz, L.E. (2021, July) *Minds at Play.* Workshop at the 43rd Annual Meeting of the Cognitive Science Society. mindsatplay.github.io

2021 **Chu, J.** & Schulz, L.E. (2021, April) *Perspectives on Play: Motivations and Constraints Across Lifespan and Species.* Symposium at the Meeting of the Society for Research in Child Development.

2019 **Chu, J.** (chair) (2019, April) *Is That So? How Children Evaluate Claims and Conjectures.* Symposium at the Bi-ennial Meeting of the Cognitive Development Society, Louisville, KY.

Editorial Service

Guest Editor, Topics in Cognitive Science, Special Issue on *Goal Dynamics In Cognition*

Journal Peer Review

Child Development, Cognitive Development, Developmental Science, Journal of Experimental Child Psychology, Journal of Experimental Psychology: General, Journal of Numerical Cognition, Nature Human Behavior, Open Mind, Topics in Cognitive Science

Conference Reviewing

Budapest CEU Conference on Cognitive Development (*BCCCD*), Cognitive Development Society (*CDS*), Cognitive Science Society (*CogSci*), International Conference on Machine Learning (*ICML*), Neural Information Processing Systems (*NeurIPS*), Society for Research in Child Development (*SRCD*)

Science Outreach

2025 Why Do Animals Play? *Outside/In & Tumble Science Podcast for Kids*

2024 Cognitive Tools, *Lynbrook High School Society for Psychology and Sociology*

2020 Ask a Scientist Panelist, *Museum of Science (Boston, MA)*

2019–2021 Lecture Coordinator & Mentor, *Science in the News (Harvard, MA)*

2019 What can babies teach us about human intelligence? *Science in the News (Harvard, MA)*

2019 What can babies teach us about human intelligence? *Beacon Hill Seminars (Boston, MA)*

2017-2018 Babies & Brains, *MIT Educational Studies Program* Half-day workshops for 7-12th grade students, co-taught with Halie Olson & Heather Kosakowski.

Professional Development

2022 Tools to Promote Culturally Responsive Mentorship, *MIT*
2022 Diversity & Inclusion Program workshops, *MIT & University of Rhode Island*
2022 Research Mentorship Workshop, *MIT*
2019 Kaufman Teaching Certificate Program, *MIT*
2018 Center for Brains, Minds, & Machines Summer Course, MIT

References

Dr. Laura E. Schulz | MIT, Department of Brain & Cognitive Sciences
email: lschulz@mit.edu
Dr. Elizabeth Bonawitz | Harvard, Graduate School of Education
email: elizabeth_bonawitz@gse.harvard.edu
Dr. Judith E. Fan | Stanford University, Department of Psychology
email: jefan@stanford.edu